the sun. This can be explained by the equation,

$$CO_2 + H_2O \rightarrow (CH_2O)_n + O_2$$

where $(CH_2O)_n$ represents the organic form of carbon (Example: Carbohydrates) which gets incorporated into the photosynthetic organisms. This organic carbon serves as food for herbivores and in turn for carnivores.

2. Decomposition

The organic matter fixed as a result of photosynthesis is eventually degraded by microorganisms to CO_2 during processes like respiration and decomposition. When aerobic and anaerobic organisms respire, CO_2 is released into the atmosphere. Much of the CO_2 is released when dead organisms decompose in the soil predominantly by the activities of soil microorganisms. Burning of fossil fuels also release CO_2 into the atmosphere.

3. Methanogenesis

It is an anaerobic process where CO_2 gets converted to CH_4 (methane) by strict anaerobes like methanogens (Example: *Methanobacterium*). Methanogens are a group of Archaebacteria found in anaerobic environments like swamps, marshes, rumen of ruminants, paddy fields and gut of termites.

$$H_2 + HCO_3^- + H^+ \rightarrow CH_4 + 3H_2O$$

Methane is converted back to carbon dioxide by a process called **Methylotrophy**.

11.1.2 Nitrogen Cycle

The element Nitrogen (N) is a key constituent in microbial cell. Nitrogen is needed for the formation of proteins, amino acids, nucleotides and is present in a number of oxidation states inside the cell. Nitrogen is cycled between atmosphere, organic compounds in living things, soil and sediments.

The processes that are involved in Nitrogen cycle are

1. Nitrogen fixation
2. Nitrification
3. Ammonification
4. Denitrification

Nitrogen fixation

Nitrogen is present as N_2 ($N \equiv N$) in air (78% N_2). The triple bonded state of nitrogen makes it very stable and nitrogen in its gaseous state cannot be assimilated by plants or animals for their metabolism. Only few groups of prokaryotes are

Infobits

Fistulated cow:

Fistulated cows are very useful in studying rumen microorganisms and ruminant nutrition and are used to treat indigestion in cows. Fistula is a sampling port that allows access to the rumen. Rumen is the first and the largest chamber of the stomach of cows and other ruminant animals.



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